

HACKATHON PROJECT · 2026

Textile MSME Multi-Agent Supply Chain

A domain-specific multi-agent system (MAS) built to make India's textile micro-clusters resilient against raw material volatility, infrastructure failures, and compliance barriers.

6

Key Visuals

4

Novel Agents

3

Network Tiers

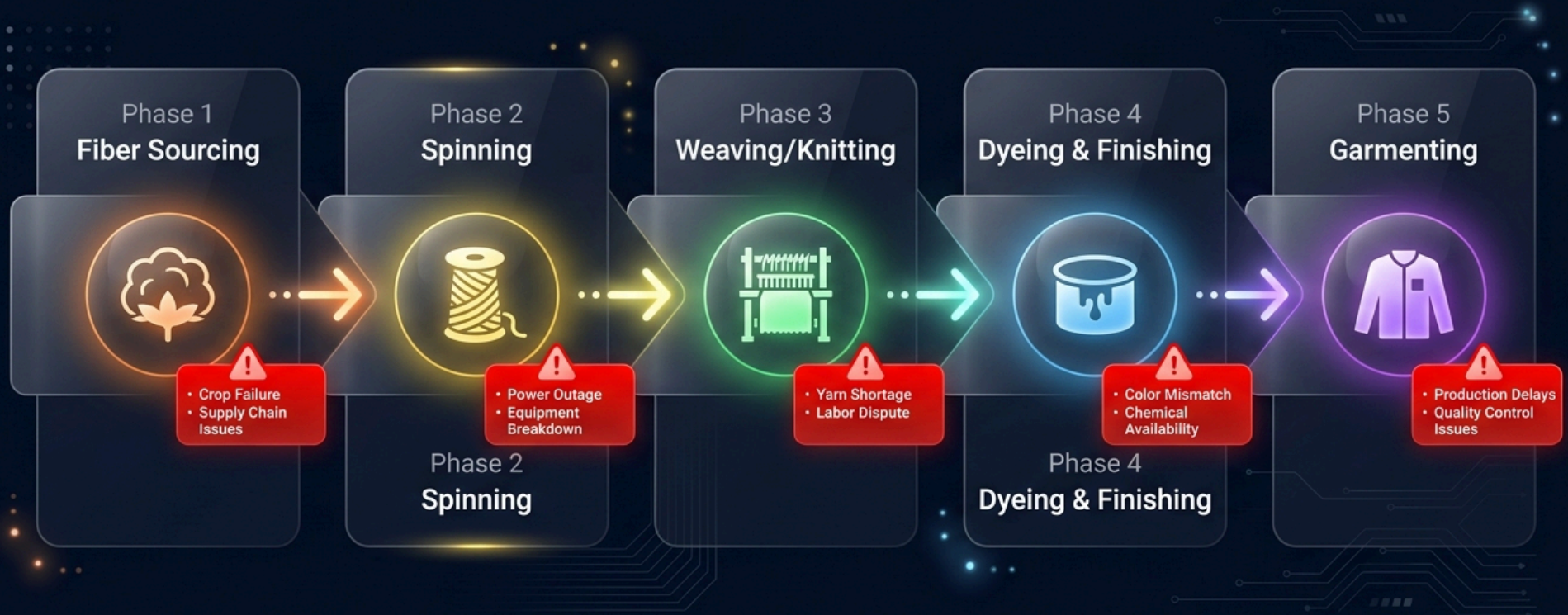
\$8

Hardware Cost

Textile Value Chain — Disruption Map

All 5 phases of textile production with their critical failure points at each stage

Textile Value Chain — Disruption Map



Our Competitive Edge — Generic vs. Domain-Specific Agents

What makes our MAS fundamentally different from every other textbook solution

From Generic to Domain-Specific: Our Competitive Edge

Generic MAS Standard Approach



Disruption Agent



Scenario Planning Agent



Rerouting & Logistics Agent



Compliance & Audit Agent

INNOVATION
GAP

Our Textile MAS



Material Degradation & Grade Agent
(predicts warp snapages)



Cluster Trust-Auction Agent
(privacy-preserving auctions)



Shade-Match & Batch Coordinator Agent
(spectrophotometer data)

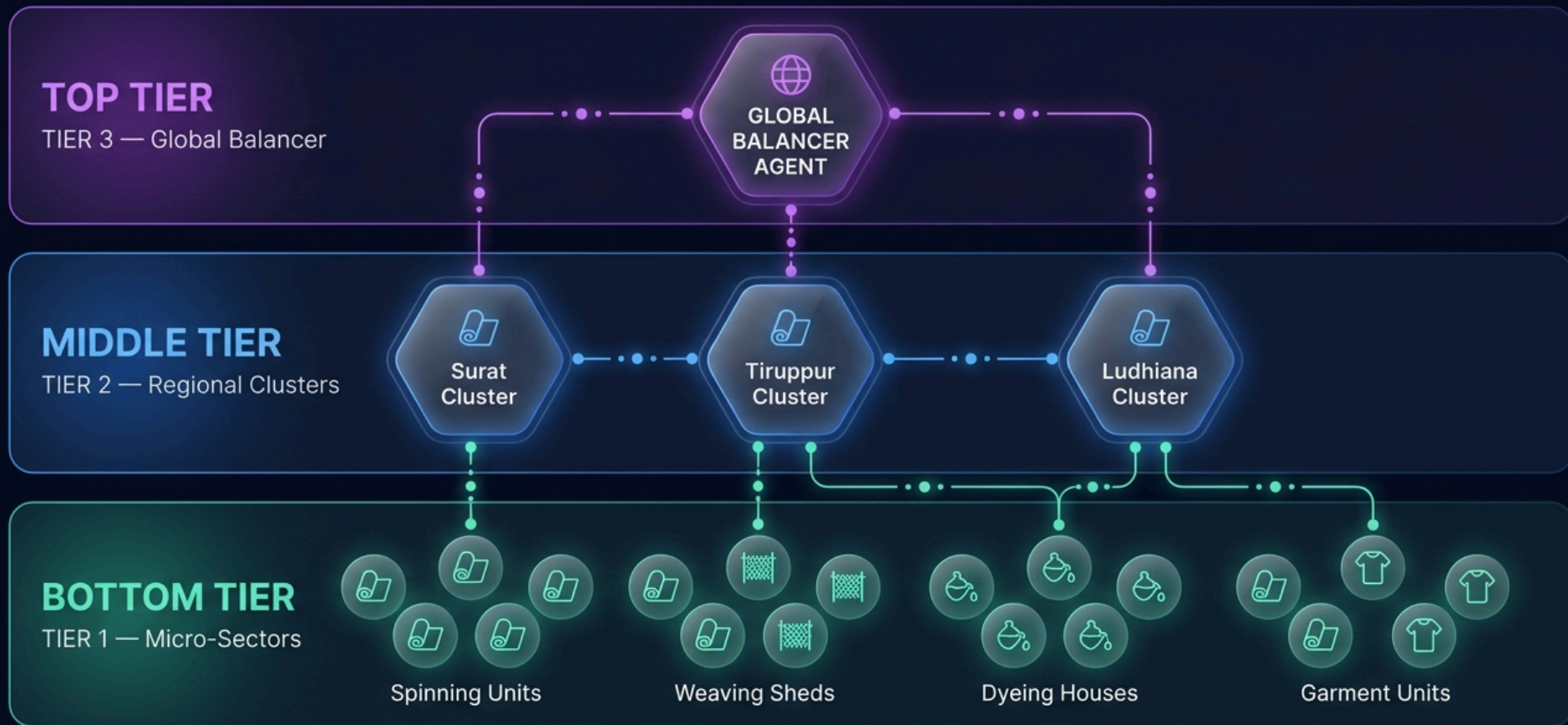


Micro-Trim Kit Checker Agent
(BOM completeness score)

3-Tier Multi-Agent Network Architecture

Micro-Sectors → Regional Clusters → Global Balancer: how the agent network is structured

3-Tier Multi-Agent Network Architecture



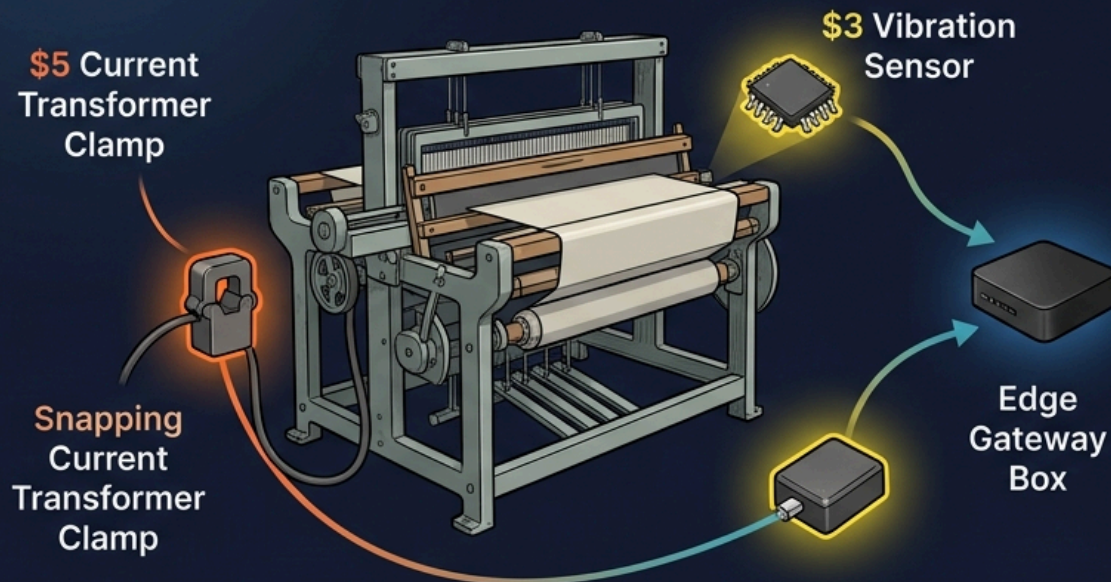
Connecting Old Looms to Smart Networks — Under \$10

The low-cost hardware bridge that connects analog machines to the token system

Connecting Old Looms to Smart Networks — Under \$10

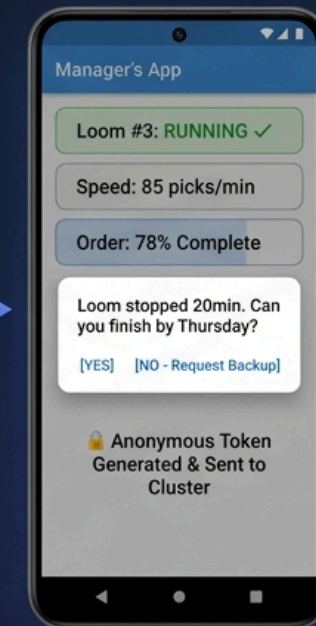
LEFT COLUMN

The Hardware



RIGHT COLUMN

The Software



No machine modification. No rewiring. Just clip, stick, and connect.

Heartbeat Protocol — Turning Silence Into Intelligence

How missed pings during a power cut automatically trigger rerouting across the cluster network

Heartbeat Protocol — Turning Silence Into Intelligence



REGIONAL POWER CUT

3 Missed Heartbeats = Failure Detected



2. Local Mesh Activated



3. Order Auto-Rerouted



Silence = Signal. Our system treats missing data as actionable information.

Zero-Knowledge Token Bidding Protocol

How supplier failure is resolved without exposing any factory's identity, price, or capacity secrets

Zero-Knowledge Token Bidding Protocol

Anonymous Bidding Token System for the textile MSME supply chain



WHY THIS WORKS

🗄️ No identity revealed

🗄️ No price leaked

🗄️ No competitor advantage

Presentation Order & Speaker Notes

#	SLIDE	KEY MESSAGE FOR JUDGES
1	Value Chain Disruption Map	Show the real-world complexity — this is not a simple logistics problem
2	Competitive Edge (Agent Comparison)	Prove your innovation gap vs. every other team's textbook solution
3	3-Tier Architecture	Show scalability and how the network mirrors real cluster trust dynamics
4	Old Looms → Smart Nodes	Kill the "but MSMEs can't afford tech" objection with the \$8 kit
5	Heartbeat Protocol	Answer the power-cut objection: "Silence is our most powerful data signal"
6	Token Bidding Protocol	Mic drop — data privacy is solved without any factory exposing their books

Judge-proof pitch line: "We don't just monitor supply chains — we model the unwritten human rules of how Indian textile clusters actually operate, from co-opetition dynamics to analog machinery and rural power grids."